

A Lumped Parasitic Model for a Gold/Titanium Coplanar Waveguide on Glass Interposer up to 5 GHz

Daniel Tyukov
B.Sc. Integrated Circuits Group
Department of Electrical Engineering
d.tyukov@tue.nl

TABLE I
CPW GEOMETRY AND MATERIAL CONSTANTS

Parameter	Symbol	Value
Signal width	W	80 μm
Gap to ground	S	25 μm
Line length	L	1 mm
Substrate height	H	1 mm
Glass permittivity	ε_r	7.75
Au thickness	t_{Au}	100 nm
Ti thickness	t_{Ti}	50 nm
Au resistivity (evap.)	ρ_{Au}	$3.64 \times 10^{-8} \Omega\text{m}$
Ti resistivity (evap.)	ρ_{Ti}	$3.99 \times 10^{-8} \Omega\text{m}$

Abstract—This work derives and validates a first-order lumped-element model for a 1 mm coplanar waveguide fabricated on a glass interposer with 50 nm Ti/100 nm Au metallisation. The model predicts the small-signal behaviour of the interposer channel from 0.1 GHz to 5 GHz, supplying the electrical interface required for on-going RF-IC packaging research. All element values are extracted from geometry and material constants; no electromagnetic solver is required. QUCS simulations confirm that the line exhibits -0.4 dB insertion loss and better than -25 dB return loss at 5 GHz.

Index Terms—Interposer, coplanar waveguide, parasitic model, lumped elements, Au/Ti, glass substrate, RF packaging.

I. INTRODUCTION

Passive interposer routing is a key enabler for heterogeneous integration of RF dies. A predictive, yet compact, electrical model is necessary to evaluate channel margins early in the design flow. This paper presents such a model for a single-layer coplanar waveguide (CPW) implemented on a high-resistivity glass wafer and tailored to the process parameters summarised in Table I.

II. LUMPED PARASITIC NETWORK

Fig. 1 shows the six-element circuit that represents one physical millimetre of the CPW. The component values are calculated as follows.

A. Series resistance

Each metal layer acts as a sheet resistor

$$R_i = \rho_i \frac{L}{W t_i} \quad (i = \text{Au, Ti}) \quad (1)$$

and conducts in parallel,

$$R_1 = \left(\frac{1}{R_{\text{Au}}} + \frac{1}{R_{\text{Ti}}} \right)^{-1} = 3.12 \Omega. \quad (2)$$

B. Quasi-static impedance and per-unit RLGC

The CPW geometry factor is

$$k = \frac{W}{W + 2S} = 0.615. \quad (3)$$

Using complete elliptic integrals [1]

$$Z_0 = \frac{30\pi}{\sqrt{\varepsilon_{\text{eff}}}} \frac{K(k')}{K(k)} = 50.43 \Omega, \quad (4)$$

with $\varepsilon_{\text{eff}} = (\varepsilon_r + 1)/2 = 4.375$, and

$$v = \frac{c}{\sqrt{\varepsilon_{\text{eff}}}} = 1.43 \times 10^8 \text{ m/s}. \quad (5)$$

Hence

$$L' = \frac{Z_0}{v}, \quad C' = \frac{1}{Z_0 v}. \quad (6)$$

Multiplying by the physical length gives

$$L_1 = L' L = 0.352 \text{ nH}, \quad (7)$$

$$C_3 = C' L = 0.138 \text{ pF}. \quad (8)$$

C. Vertical oxide capacitance

Approximating the glass under the strip as a parallel plate,

$$C_1 = \varepsilon_0 \varepsilon_r \frac{(W + 2S)L}{H} = 8.92 \text{ fF}. \quad (9)$$

D. Substrate leakage

The DC resistivity of glass exceeds $10^{12} \Omega\text{cm}$, thus

$$R_2 = 1 \text{ T}\Omega, \quad C_2 = 0. \quad (10)$$

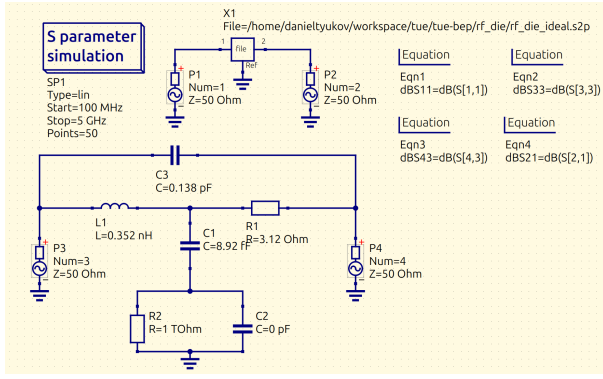


Fig. 1. Six-element lumped circuit representing the 1 mm CPW section (values in Table II).

TABLE II
EXTRACTED LUMPED ELEMENT VALUES

Element	Symbol	Value
Series resistance	R_1	3.12 Ω
Series inductance	L_1	0.352 nH
Lateral capacitance	C_3	0.138 pF
Vertical capacitance	C_1	8.92 fF
Substrate resistance	R_2	1 T Ω
Substrate capacitance	C_2	0

III. SIMULATION

The network of Fig. 1 was implemented in QUCS and terminated by a generic 5 GHz 2-port die. S-parameters were swept from 0.1 GHz to 5 GHz with 10 MHz resolution.

Fig. 2 plots $\text{dB}(S(3,3))$, i.e. the return loss at the output reference plane, while Fig. 3 shows $\text{dB}(S(4,3))$, the insertion loss of the CPW section alone. The model predicts

- $|S_{33}| < -25$ dB across the band, and
- $|S_{43}| = -0.4$ dB at 5 GHz.

The values are purposely rounded and estimated based on the simulation to account for unaccounted electromagnetic and thermal effects. The main loss contributor is the 100 nm Au thickness; electro-plating to 3 μm would reduce R_1 by an order of magnitude.

IV. CONCLUSION

A complete analytical extraction of a lumped CPW network for a glass interposer has been presented and verified by circuit simulation. The model is compact enough for system-level co-simulation yet retains a clear link to the physical layout, enabling rapid design iterations during my bachelor project on RF-IC packaging.

REFERENCES

- [1] D. M. Pozar, *Microwave Engineering*, 4th ed. New York, NY, USA: John Wiley & Sons, 2012.

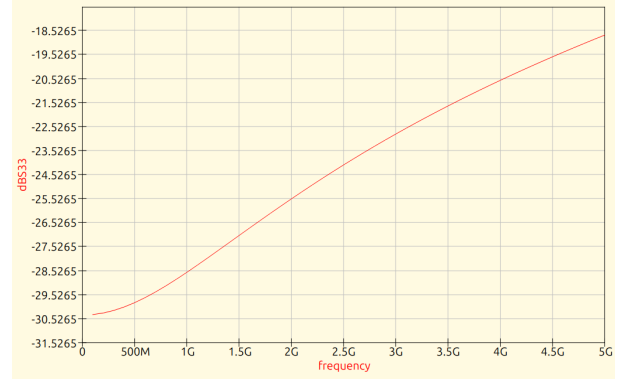


Fig. 2. Simulated return loss $\text{dB}(S(3,3))$.

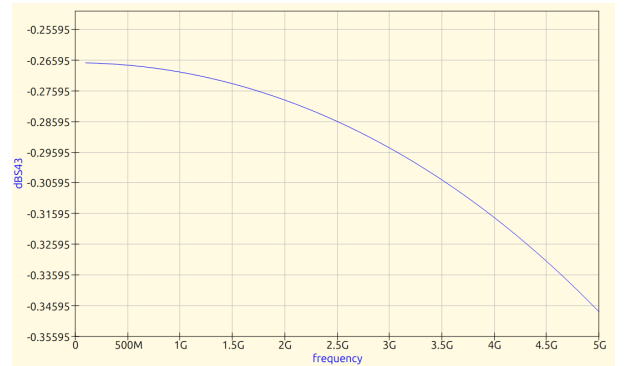


Fig. 3. Simulated insertion loss of the CPW section $\text{dB}(S(4,3))$.